Tristan McCaskill

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Education

University of Michigan — Ann Arbor, MI
B.S.E. in Computer Science, College of Engineering — *Aug 2024 – Expected May 2028*

- **Relevant Coursework:** Elementary Programming Concepts, Data Structures & Algorithms, Computer Organization, Foundations of Computer Science, Cryptography, Discrete Mathematics, Linear Algebra

Experience

Programming Director (prev. Research and Development Lead) — Jan 2026 – Present
WolverineSoft (~20-member student game studio): File Not Found — *Ann Arbor, MI*

- Led the R&D prototype of a cyber-themed Unity roguelike in a full-cycle production course, managing two artists, a sound designer, and a programmer while providing technical direction.
- Programmed procedural floors from hand-authored isometric room prefabs, a coroutine-driven floor loop, and data-driven upgrades built from paired ScriptableObject assets with reversible effects.
- As programming director, reviewed and merged 34 merge requests in one semester covering weapons, saving, inventory, enemy pathfinding, bosses, and UI, resolving conflicts to keep `main` stable.
- Refactored a monolithic upgrade runtime into interface-based behaviors, added a central UI manager and pooled audio, and maintained the project's 35-page developer wiki.

Studio Programmer — Aug 2025 – Aug 2026
WolverineSoft: Project Witches, a farming and automation game — *Ann Arbor, MI*

- Built the Flowvine item network: breadth-first search groups connected plants into systems and Dijkstra's algorithm routes each item to the fastest output, returning items safely if a route is cut.
- Replaced a hard-coded summer/winter flag with an event-driven season system, and resolved bugs in the in-game weather cycle.
- Implemented hydration, harvest chance, and timed plant spells as components with reversible modifiers; extended saving to persist multiple concurrent quest lines.
- Built late-game content including the Raphael Lotus win condition and a water-walking Frostberry that toggles tile colliders around the player.

Grader, EECS 298: 3D Technical Art and Animation — Aug 2025 – Dec 2025
University of Michigan — *Ann Arbor, MI*

- Graded Blender models, textures, and animations for 80+ students, writing rubric-based feedback on each submission.
- Assessed Unity integration projects and 3D asset portfolios for technical and artistic accuracy.
- Clarified assignment guidelines, answered student questions, and tracked submissions in GitLab and Canvas.

Projects

Cachi: Multi-level Cache Simulator | *C, Make, Valgrind* — GitHub

- Simulates configurable cache hierarchies with per-level capacity, block size, associativity, and LRU, FIFO, MRU, or random replacement, reporting hits, misses, evictions, and writebacks per level.
- Replays memory traces from Valgrind's Lackey tool, captured from custom C benchmark programs.

Canonical Huffman File Compressor | *C, Make, Python* — GitHub

- Command-line compressor using canonical, length-limited (15-bit) Huffman codes; reduced the Canterbury Corpus to 46.8% of its original size, benchmarked against `gzip -9` and analyzed performance with Matplotlib.

Awards & Game Jams

- **1st Place**, Bluebird Studios / MGDC Game Jam: *Protect Your Data*, $750 prize — Mar 2025
- **1st Place (Local Site)**, WolverineSoft / Global Game Jam: *Bubble Casino* — Feb 2025
- **Final Showcase Winner**, EECS 183 Elevator Algorithm Project — Dec 2024
- **3rd Place**, WolverineSoft Turkey Jam: *Don't Flock Up*, an asymmetric PvP runner — 2024
- **3rd Place**, WolverineSoft Pumpkin Jam: *Block Party*, a rhythm game — Oct 2024

Technical Skills

Languages: C, C++, C#, Python, HTML/CSS, JavaScript, LaTeX
Tools & Engines: Unity, Roblox Studio, Blender, Git, GitLab, Valgrind, Make, Matplotlib, Linux